

Writing Large Numbers in Scientific Notation

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Lesson 8 — Writing Large Numbers in Scientific Notation	Date	
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.4

2. Standards

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Must. Express very large quantities in scientific notation and read them back into standard form; determine how many times larger one number is than another.

Must not. Coefficients are at least 1 and less than 10. Large numbers today; negative powers of ten are 3.09, and operations in scientific notation are Unit 4.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR. 7.1	Apply mathematics to real-world contexts.	Every board and the bank — populations, distances and counts from outside the classroom, written the way science writes them.
MA.K12.MTR. 5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the powers of ten are set against place value, so the exponent is read as a count of places rather than a count of zeros.
MA.K12.MTR. 6.1	Assess the reasonableness of solutions.	Board 4 — the comparison is settled by the exponent first; students say which number is larger before they look at the coefficients.

4. Learning Target and Essential Question

Learning target. I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

Essential question. How does a power of 10 let us write a very large number briefly, and compare two of them?

Building on. MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.07) — powers of 10 and the laws; Grade 5 — place value to the billions.

Working toward. 3.09 — small numbers and negative powers of ten; MA.8.NSO.1.5 (Unit 4) — operations in scientific notation.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the 'today' line and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
9.5×10^5 has the larger coefficient but is the smaller number. Explain how that happens.	DOK 2
Marcus writes 20×10^5 . It equals 2,000,000 but it is not scientific notation. What work is the coefficient rule doing?	DOK 3
Two numbers are each written as a coefficient times a power of ten. When can you compare the coefficients, and when can you not?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Write 4,000,000 in scientific notation.	4×10^6	4×10^7 — counted the digits instead of the places the point moves [B1G-M 8.NSO.1.4: exponent = places moved]; 40×10^5 — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]
2	Write 7.2×10^4 in standard form.	72,000	720,000 — the point was moved five places [8.NSO.1.4: 10^4 moves the point four places]; 7.20000 — zeros appended without moving the point
3	A stadium's renovation used 356,000 bolts. The stadium seats 71,000 people. Write the number of bolts in scientific notation.	3.56×10^5	356×10^3 — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception]; 3.56×10^3 — counted the zeros instead of the places the point moves; 7.1×10^4 — the wrong number was converted
4	Which is greater, 9.5×10^5 or 1.2×10^6 ?	1.2×10^6	9.5×10^5 — compared the coefficients and ignored the powers [B1G-M 8.NSO.1.4: order of magnitude first]
5	Which is 52,000,000 written in scientific notation?	A — 5.2×10^7	(B) equal in value, but the coefficient 52 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]; (C) counted the six zeros instead of the seven places the point moves [8.NSO.1.4: exponent = places moved]; (D) counted the eight digits [8.NSO.1.4: exponent = places moved]
6	How many times larger is 6×10^8 than 6×10^5 ?	A — 1,000	(B) 10^3 times read as 3 times [B1G-M 8.NSO.1.4 misconception: ' 10^4 times' read as '4 times']; (C) 3×100 — the exponent difference multiplied by a stray 100 [8.NSO.1.4: the ratio of the powers is 10^3]; (D) $8 \div 5$ — the exponents were divided [8.NSO.1.3: quotient of powers subtracts the exponents]
7	How many times larger is 8.4×10^6 than 2.1×10^4 ?	400	4 — coefficients only [8.NSO.1.4: both factors are compared]; 40 — 10^2 read as 10 [B1G-M 8.NSO.1.4 misconception: exponent read as the multiplier]
8	A computer displays a number as 3.5E9. Write the number in standard form.	3,500,000,000	'error' or 3.5^9 — E read as something other than $\times 10^9$ [B1G-M 8.NSO.1.4 misconception: calculator E notation]; 35,000,000,000 — point moved ten places
9	Marcus writes 2,000,000 as 20×10^5 . Is his expression equal to 2,000,000? Is it scientific notation? Explain.	Equal, yes. Scientific notation, no: 20 is not less than 10. 2×10^6 .	'not equal' — value confused with form [8.NSO.1.4: equivalent expressions can differ in form]; 'yes, it is scientific notation' — the coefficient constraint was not applied [B1G-M 8.NSO.1.4 misconception]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the place-value strip from Notes I — 10^0 through 10^9 with the place names under it — and let the student walk the decimal point along the strip, one place per power, instead of counting zeros in their head.

ELL. Coefficient and standard form are the two phrases the questions use. Write $3.56 \times 10^5 = 356,000$ once with COEFFICIENT under the 3.56 and STANDARD FORM under the 356,000, and point at them all period.

Enrichment. Ask how many times larger 6.4×10^9 is than 1.6×10^5 , then for a number whose scientific notation has a coefficient of exactly 1.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say both things — equal in value, and not in scientific notation — and give 2×10^6 . Read the boards, not the papers.

Vocabulary. **scientific notation; standard form; coefficient**

Materials. Whiteboards and markers. Calculators are allowed on everything; today is the day to show the class what E means on the display — enter 3.5×10^9 and 35,000,000,000 and look.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Convert between standard and scientific notation: large numbers — 2AL; Compare large numbers — WEZ; Also consider: Powers of ten — LJQ.